Milestone 4: Model Training Report

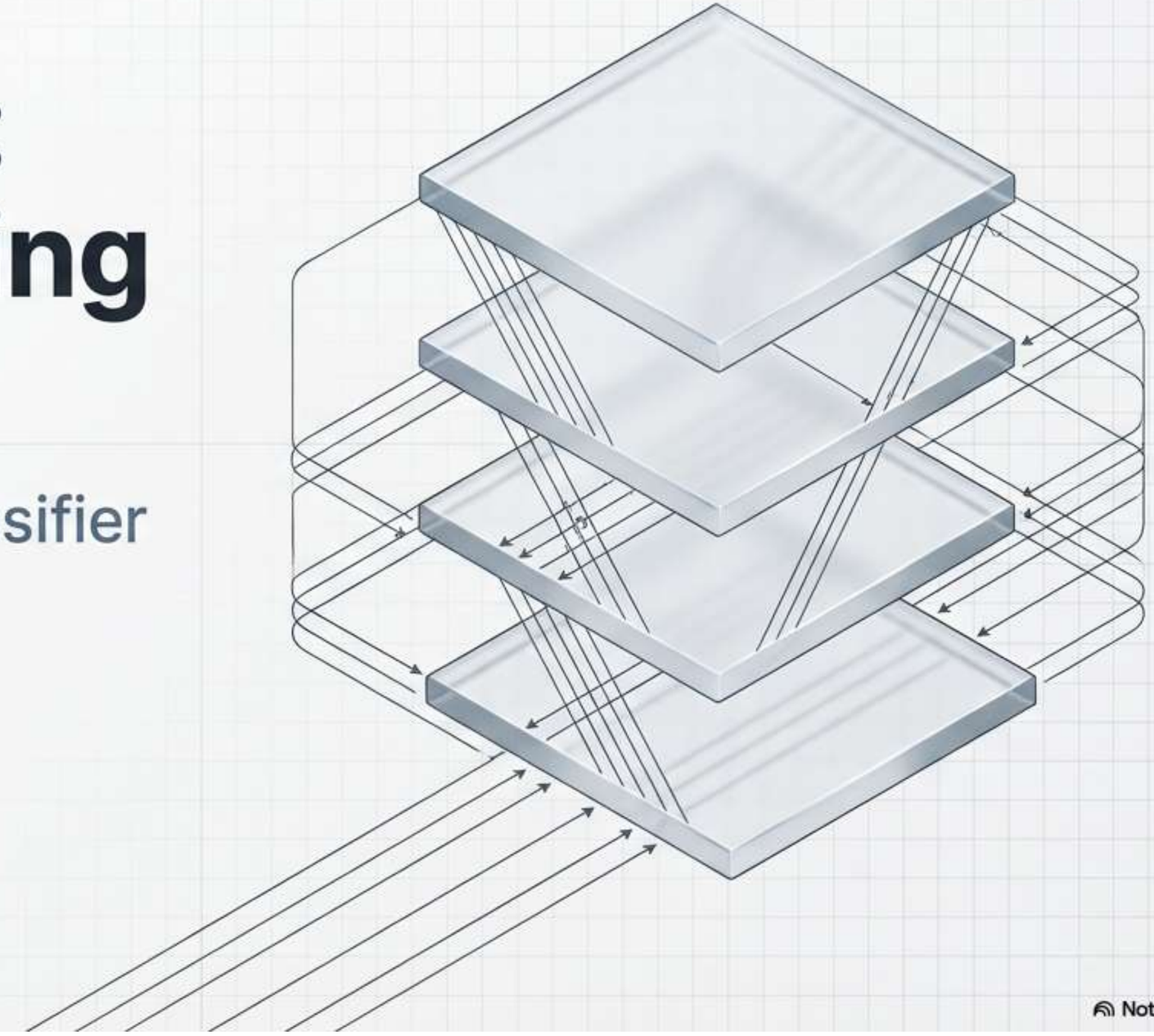
Transformer Fraud Classifier

RoBERTa-base

Optuna HPO

Focal Loss

~125.5M Parameters



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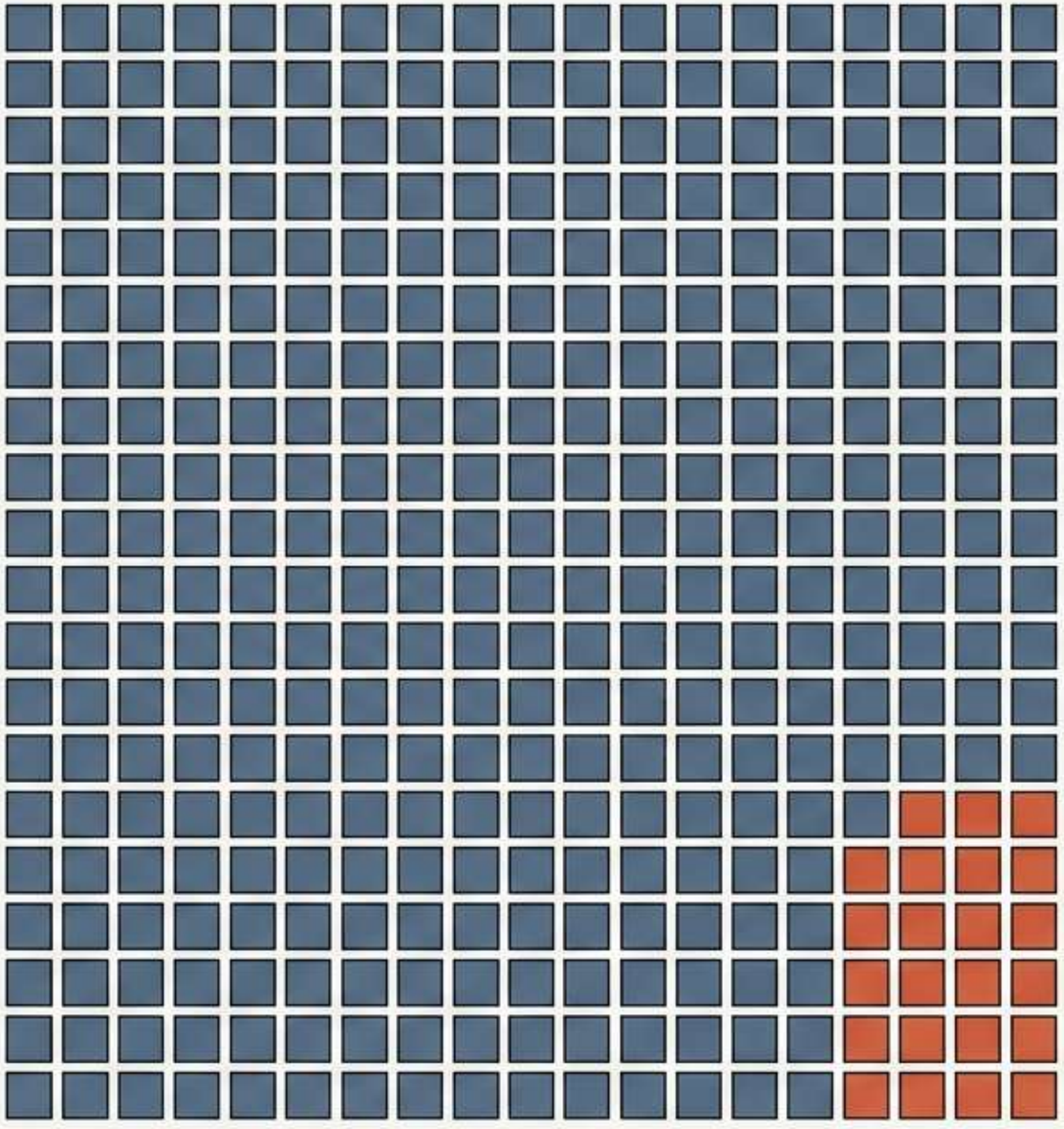
The Objective and the 20:1 Antagonist

Goal: Build an AI model to automatically detect fraudulent job postings.

Data Inputs (Unstructured)
Title, Description, Requirements, Company Profile, Benefits.

Data Inputs (Structured)
Location, Department, Salary Range, Employment Type, Industry, Logo Presence.



17,880
Total Listings

95.2%
Legitimate

4.8%
Fraudulent

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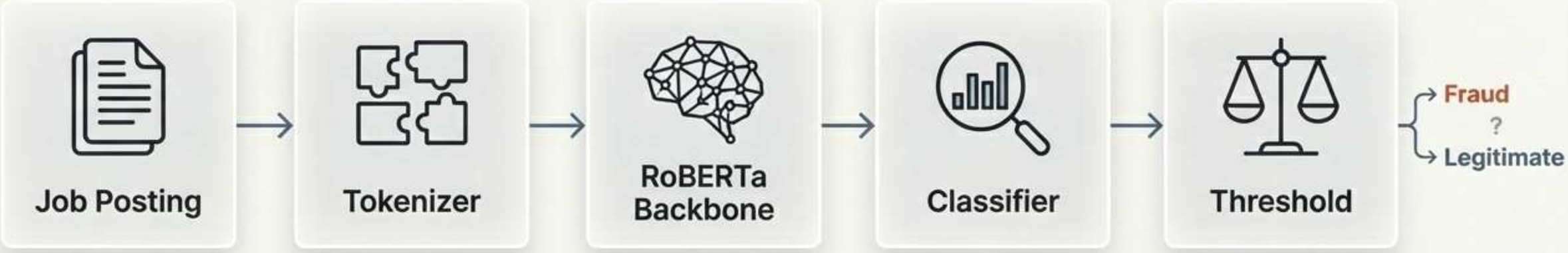
Unpacking the RoBERTa Pipeline



Text & Structured Data

Splits into word-pieces (Max 512 tokens)

Reads and contextualises the text

Outputs continuous fraud probability (0 to 1)

Binary decision: Fraud or Legitimate?

Model Size at a Glance	
Total Parameters:	~125.5M
Transformer Layers:	12
Hidden Dimensions:	768

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Engine Room Configuration and Telemetry

01

Loss & Optimisation

Loss Function: Focal Loss (Focuses on hard-to-classify examples).

Optimiser: AdamW.

Learning Rate: ~`2e-5 to 3e-5` (Automatically tuned by Optuna).

02

Batching & Epochs

Batch Size: 16 (Effective 32 via gradient accumulation).

Epochs: 8–13.

Regularisation: Early stopping applied after 5 non-improving rounds.

03

Hardware Constraints

Compute: Google Colab T4 GPU.

Precision: Mixed precision (FP16) for processing speed.

04

Stability Metrics

Gradient Clipping: `max_norm = 1.0` (Prevents unstable training updates).

Evaluation Metric: F1-score (fraud class) targeted at `≥ 0.91`.

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The Evolutionary Matrix

v1 Baseline

RoBERTa + basic weighted loss. No auto-tuning.

v2 Focal Loss

Switch to Focal Loss; improved spotting of rare fraud.

v3 Auto-Tuning

First automated Optuna search (15 trials).

v3_1 (FINAL)

25 Optuna trials.

Dynamic γ (gamma) + hard precision/recall floors.

The Winning Model.

v4 DeBERTa:

Alternative model backbone tested. Slower, with no clear advantage.

v5 & v5s:

Cosine LR & Synthetic data. AI-generated fake jobs require further filtering.

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The Strategic Arsenal

Focal Loss & Class Weighting

Mathematically forces the model to focus on difficult, rare fraud examples.

Penalises the model 20× more for missing a fraudulent post than for a false alarm.

Optuna HPO

Replaces manual guesswork with an automated `25-trial` search algorithm.

Hunts for the optimal combination of learning rates and dynamic gamma values.

Threshold Tuning

Abandons the default 50% probability cut-off.

Dynamically locates the perfect mathematical boundary to balance precision against recall.

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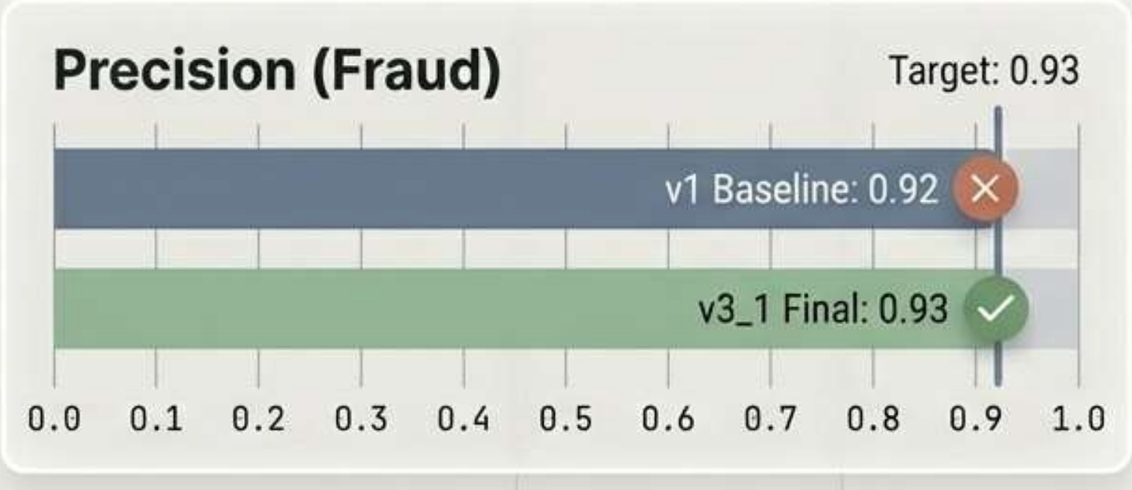
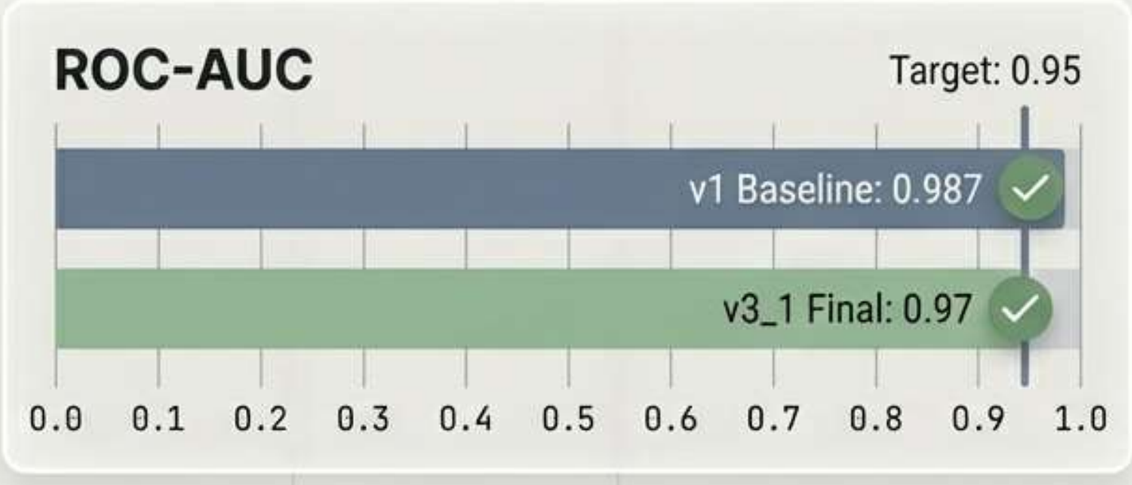
Surpassing the Performance Baselines

Context Note: Targets established by Mahfouz et al. (2019).





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Scientific Retrospective

What Worked ↓	What Did Not Work ↓
Focal Loss Dramatically outperformed standard weighted loss (+3–5% recall).	**Layer-wise LR Decay** Introduced pipeline complexity but yielded negligible benefits.
Optuna (25 Trials) Discovered hyperparameter settings that manual searches completely missed.	**DeBERTa (v4)** Slower tokenisation with no distinct accuracy advantage over RoBERTa.
Threshold Tuning Generated a “free” 2–4% F1 gain with zero additional training cost.	**Extreme Focal Gamma** Setting $\gamma=3.0$ proved far too aggressive, immediately degrading precision.
Full Fine-Tuning Proved superior to LoRA adapters for a dataset of this specific size.	**Raw Synthetic Data** AI-generated fake jobs show promise but require strict quality filtering to be viable.

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Outstanding Operational Friction

Imbalanced Data

The constant `20:1` ratio means the model maintains an ongoing, fragile reliance on Focal Loss and precise class weighting to avoid defaulting to a 95% “everything is legitimate” baseline.

Compute Limits

Free-tier Google Colab T4 GPUs heavily restricted epoch counts (~10 per run) and capped the total number of Optuna trials possible.

Token Truncation

~11% of job postings exceed RoBERTa’s strict 512-word limit, meaning critical fraud signals buried at the bottom of long descriptions are currently ignored.

The Precision vs. Recall Tension

Simultaneously demanding ≥93% precision and ≥89% recall creates a mathematical tightrope; optimising one metric inherently degrades the other.

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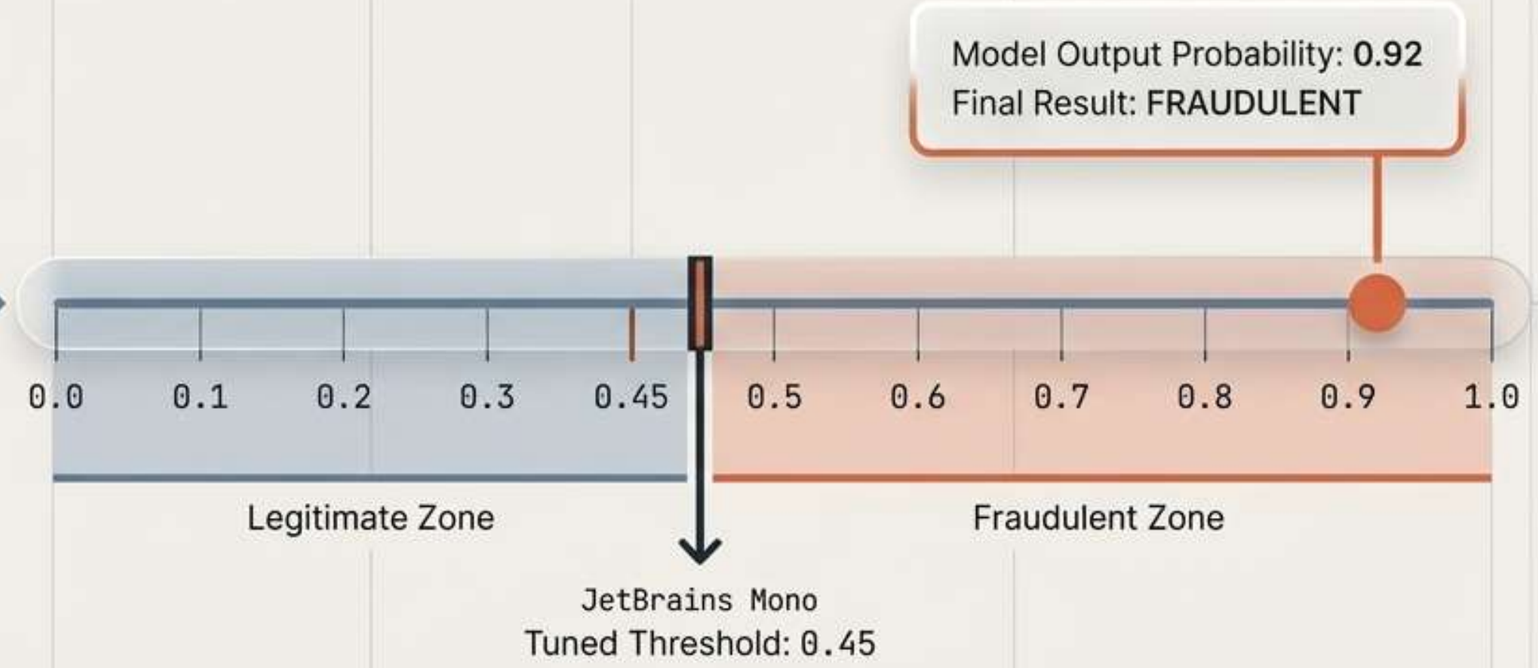
Dynamic Tension in Decision Making

The probability threshold is precisely tuned on validation data to maximise the capture of fraud while aggressively limiting false alarms.

Input Card

Work From Home Data Entry Specialist... Earn $500/day working from home! No experience needed.

Salary: $500–$1000/day | Company Logo: No



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Key Technical Takeaways

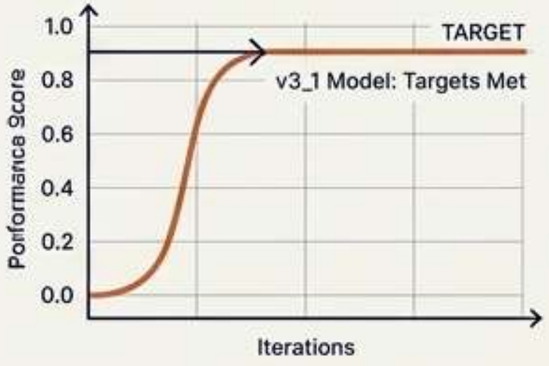
Targets Achieved: The final `v3_1` model successfully conquered the `20:1` imbalance, securing `F1 ≥ 0.91`, `Recall ≥ 0.89`, `Precision ≥ 0.93`, and `AUC ≥ 0.97`.

The Winning Formula: The exact combination of **Focal Loss** + **Optuna HPO** + **Dynamic Threshold Tuning** was required to crack the dataset.

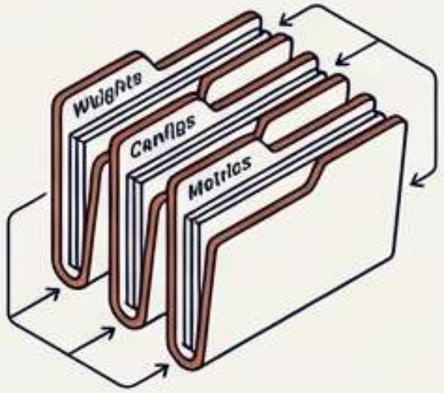
Next Phase: Development is now shifting from iterative training to packaging the model as a live **API endpoint** for real-world deployment.

Milestone 4: Deliverables & Team Output

Rigorous Targets Achieved



The final v3_1 model successfully met all stringent F1, Precision, and Recall targets set for Milestone 4.



Comprehensive Artefact Management

Strict versioning of output artefacts.
Documentation includes plotted training curves and deep qualitative observations.

Seamless Collaboration

Every phase—from deep learning architecture to agentic generative AI integration—was powered by joint architectural discussions, regular code reviews, and unified engineering.

ARUN DUTTA
HRITIK ROSHAN MAURYA
VIVEK BAJAJ
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